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GEOLOGY

Paleomagnetism of Ordovician–Silurian Volcanics on the Western Slope of the Southern Urals

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Abstract—This work presents new paleomagnetic data on previously dated Ordovician–Silurian volcanics from four sections in the western framework of the Taratash massif (Southern Urals). The geological data indicate that the region under study was a part of the Baltica paleocontinent since the early Mesoproterozoic. The paleomagnetic data obtained for rocks of the westernmost part of the Ural Fold Belt show no evidence of local and regional rotations relative to Baltica. Consequently, the pole obtained can be extrapolated to the entire platform. The result obtained is characterized by a rather high degree of reliability and can clarify the trajectory of the apparent polar wander path (APWP) for the Baltica paleocontinent during the Late Ordovician–Early Silurian, where reliable paleomagnetic data are not available, and can be used for further paleoreconstructions.

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In recent years, using U/Pb zircon dating (SHRIMP and ID-TIMS) of the Riphean igneous formations in the Southern Urals, Paleozoic concordant ages (437 ± 11 and 441.8 ± 8.2 Ma [1]) were obtained for basalts, which lie among volcanogenic–sedimentary deposits of the Lower Riphean Ai Formation. Basalts are closely connected with both the Riphean volcanic rocks and sedimentary rocks of the Ai Formation [2]. The paleomagnetic study of Late Ordovician–Silurian volcanics is of particular interest in that it provides us a unique possibility to acquire new data and to clarify the trajectory of the apparent polar wander path (APWP) for the Baltica paleocontinent during the Late Ordovician–Early Silurian. The paleomagnetic data obtained can be used for further paleoreconstructions. A review of the paleomagnetic data available for the earlier Vendian–Early Ordovician segment of the APWP is given, for example, in works [3–5]; the later segment of the APWP is characterized in [4]. There are no reliable paleomagnetic data for the East European Platform in the interval 432–458 Ma, and on the pole migration curve constructed for the Baltica paleocontinent [4], only a calculated long line without the actual data is shown.

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Based on the geological data, the western segment of the Ural Fold Belt is considered to be the deformed margin of the platform [6, 7]. It was shown previously based on the paleomagnetic data that there was no significant displacement of the westernmost part of the Southern Urals relative to the platform [8, 9]. In addition, local turnouts of separate tectonic blocks relative to each other were not revealed as well. Accordingly, the pole obtained on Ordovician–Silurian rocks of the westernmost segment of the Urals can be extrapolated to the entire platform. This work presents new paleomagnetic data obtained for the previously dated Ordovician–Silurian volcanics from four sections located on the western framework of the Taratash massif (Fig. 1).

Sections are composed of lava flows of Paleozoic subalkaline basalts with a low degree of secondary alterations. These flows are in the field of the Lower Riphean volcanics of the Ai Formation. The concordant dates of zircons from igneous rocks of the studied sections are confined to an interval of 437–444 Ma [1, 2]. In total, 18 lava flows were distinguished in sections exposed in basins of the Ushat and Shmelevka rivers, to the northeast of the village of Arshinka, and on Mt. Malyi Miass. The dip and strike (dip azimuth/dip angle) of lava flows were measured based on fluidization and/or bedding of country sedimentary rocks: 305°–310°/45°–57° in the Ushat and Arshinka sections, 75°/70° in the Shmelevka section, and 240°/40° in the Malyi Miass section (Fig. 1). For paleomagnetic study, 7–13 samples were collected from every flow

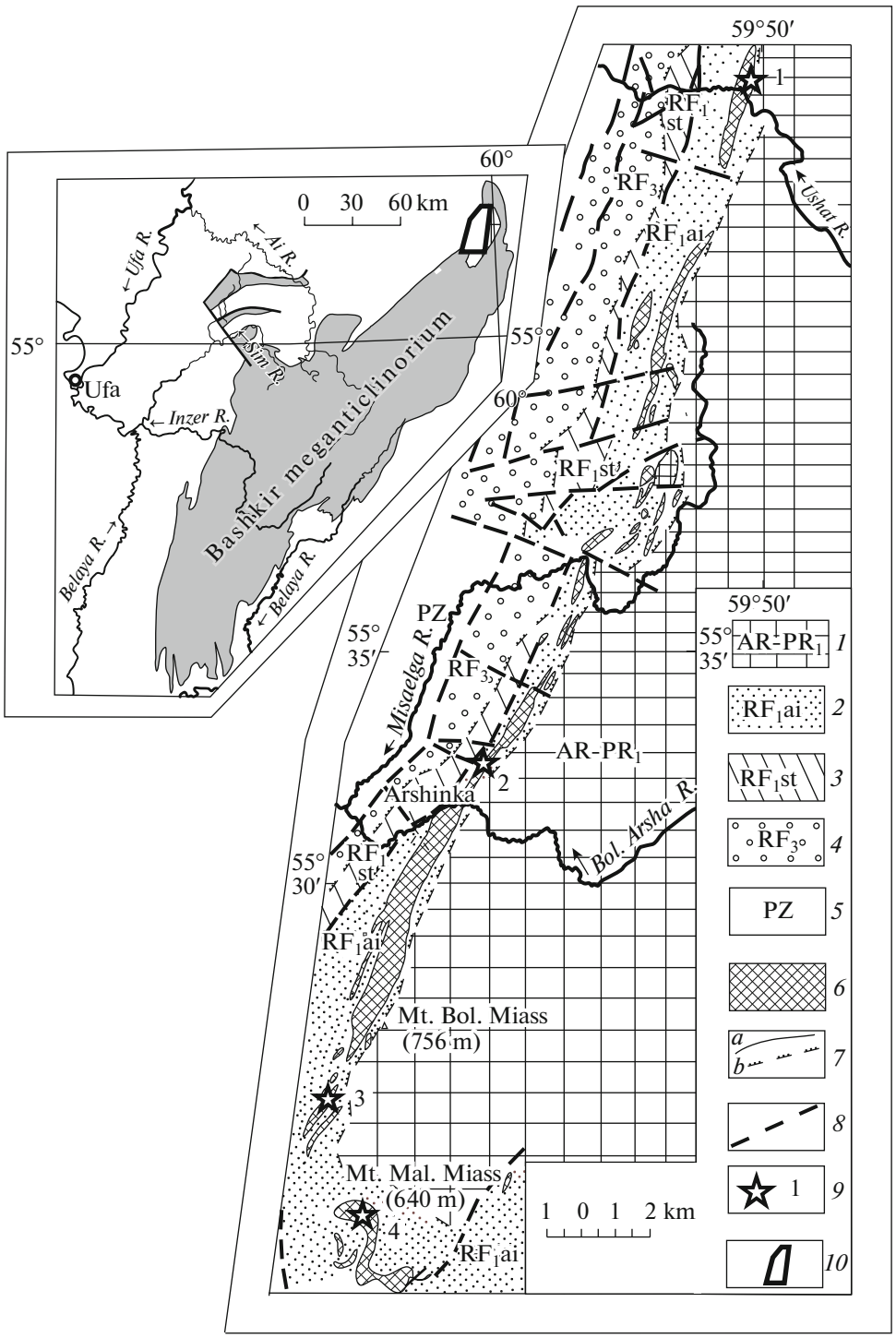


Fig. 1. Schematic geological map of the distribution of volcanics of the Ai Formation (Taratash anticlinorium, Southern Urals) with an indication of sampling sites (after [2] with amendments). (*1*) Archaean–Early Proterozoic formations of the Taratash Complex (AR–PR_1); (*2–3*) Lower Riphean deposits: (*2*) Ai Formation: (R_1ai) and (*3*) Satka Formation (RF_1st); (*4–5*) undivided deposits: (*4*) Upper Riphean (RF_3) and (*5*) Paleozoic (Pz); (*6*) volcanic rocks; (*7*) geological boundaries: (*a*) concordant and (*b*) discordant; (*8*) tectonic contacts; (*9*) sampling sites and their numbers (1, Ushat R., nine flows; 2, northeast of the Arshinka village, one flow; 3, Shmelevka R., four flows; 4, Mt. Malyi Miass, four flows); (*10*) area of study (overview scheme).

and united into separate sites for data processing. In total, 170 samples were studied. The data on 115 samples from all sites were used for further interpretation.

The laboratory studies were performed in accordance with the currently accepted technique. All samples were subject to detailed temperature cleaning.

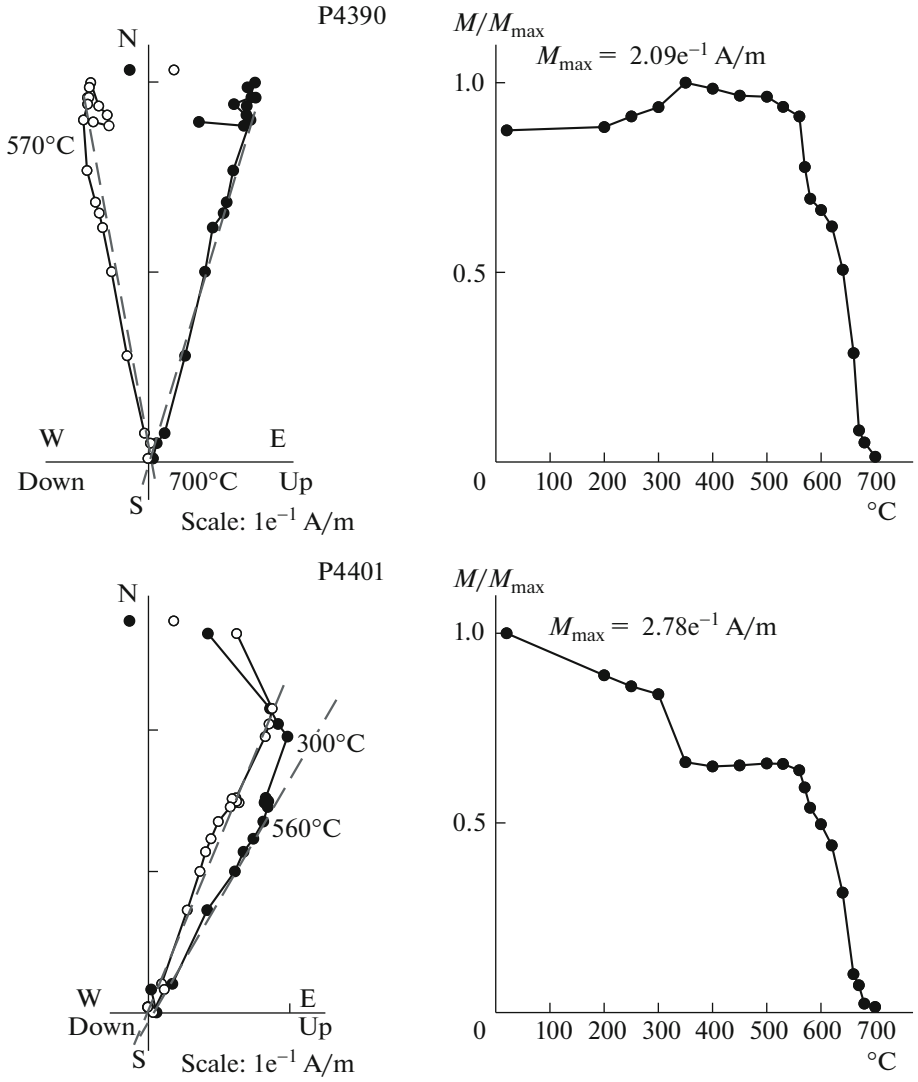


Fig. 2. Results of paleomagnetic studies. Zijderveld diagrams in the stratigraphic system of coordinates and curves of thermal demagnetization of some samples. Magnetization components are shown by dashed lines. Open symbols are the projection of natural remanent magnetization J_n on the vertical plane; solid symbols are those on the horizontal plane.

The residual magnetization was measured with the JR-6 spinner magnetometer.

All rock samples are characterized by strong magnetic signals. The natural residual magnetization varies in the range of 0.1–8.0 A/m (2.0 A/m, on average) reaching 200 A/m in single samples. As seen in the stereograms, remagnetization circles can be distinguished for some samples in the temperature interval from 250–300 to 480–600°C. The high-temperature characteristic component can be distinguished in a narrow interval from 560–580 to 680–700°C and in a wide one from 250 to 300°C up to demagnetization (Fig. 2). As usual, the component is directed towards the origin of coordinates. In fact, there are no signs of Late Paleozoic remagnetization in the samples studied. As follows from the step by step thermal demagnetization curves, hematite and magnetite are magnetization carriers. The distinguished high-temperature magnetization components are unipolar (Fig. 3). Due to this, we were not able to perform a reversal test. However, the result obtained is in agreement with the suggestion on the existence of the normal polarity superchrone in the Late Ordovician–Early Silurian [10]. The direction–correction (DC) tilt test [11] performed for the folds yielded a positive result. The distinguished mean direction of the high-temperature component corresponds to the paleolatitude $-1.1° \pm 4.0°$. The coordinates of the calculated paleomagnetic pole (eighteen sites) are 25.6° N, 197.2° E (25.6° S, 17.2° E), and the radius of the confidence circle near

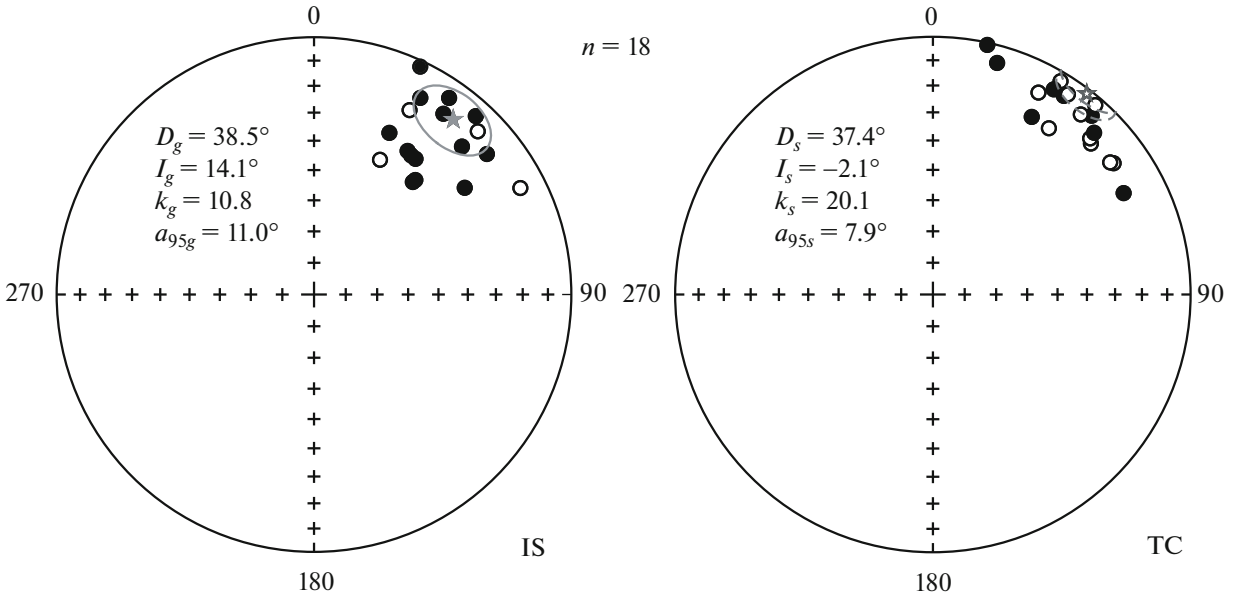


Fig. 3. Directions of high-temperature magnetization components of the studied sites in geographic (g) and stratigraphic (s) systems of coordinates. Solid symbols are projections of the magnetization vector on the lower hemisphere; open symbols are those on the upper hemisphere. A star shows the general mean direction and its confidence ellipse.

the pole A_{95} = 6.2° is in good agreement with the calculated data obtained for the Baltica paleocontinent (Fig. 4). The positive fold test and consistency with the trajectory of the apparent polar wander path for the Baltica paleocontinent make it possible to suggest that the distinguished magnetization component can be regarded as the primary one.

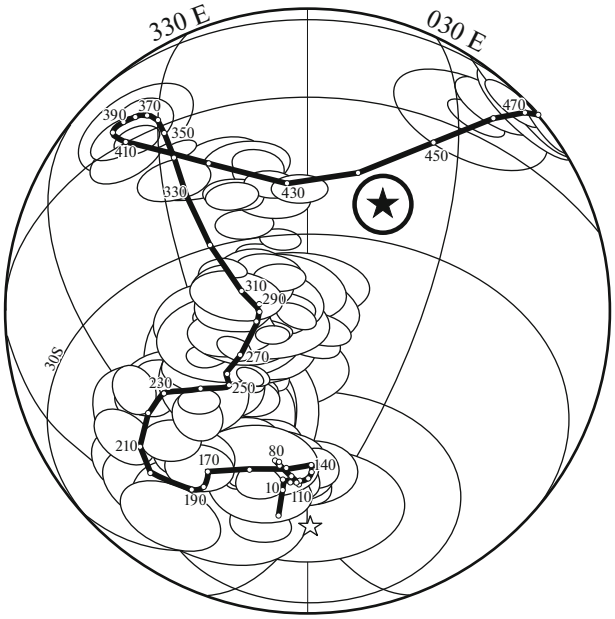


Fig. 4. The position of the calculated average paleomagnetic pole (a star) with a confidence ellipse for the studied rocks (thick) in comparison with the curve of apparent migration of the Baltica pole according to [4]. The confidence ellipses (thin) are shown for the actual data used in [4] for constructing the trajectory of apparent polar wander path.

Thus, the U/Pb zircon dating (SHRIMP) of the studied volcanics yielded an age of 437–444 Ma. Using detailed temperature cleaning, the high-temperature magnetization component was distinguished for a large number of samples from 18 sites (four sections). It is evident that this magnetization is due to the presence of hematite and magnetite. The suggestion of the primary nature of the magnetization is confirmed by the fold test and regional concordance test. The geological data previously obtained testify that the studied region is part of the Baltica paleocontinent from the early Mesoproterozoic. The paleomagnetic data obtained on the rocks of the westernmost part of the Ural Fold Belt did not reveal local and regional rotations relative to the Baltica paleocontinent. The pole obtained is inconsistent with any later poles. We assume that the new data obtained have a rather high degree of reliability and could complement the APWP trajectory for the Baltica paleocontinent at the Late Ordovician–Early Silurian segment; in addition, they can be used for paleoreconstructions.

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